

Global Companies Financial Performance Analysis 2024

Task 04: Complete Business Analytics Project

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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

1. Introduction

The aim of this project is to take a raw business dataset and turn it into something that supports decisions. The dataset used here lists 2,001 of the world's largest companies in 2024 with their country, sales, profit, assets and market value. The analysis follows the stages in the task: data preparation, exploratory analysis, KPI development, insights and recommendations, and a machine learning model as the bonus part.

All money values in this report are in US dollars and are shown in billions unless stated otherwise.

2. Data Preparation

The raw file had seven columns. The first one, "Unnamed: 0", was only an index left over from saving the file, so it was dropped. Sales, Profit, Assets and Market Value were stored as text such as "\$252.9 B" or "\$590 M", so they could not be used for calculations.

These four columns were converted to numbers in billions of dollars. The dollar sign and commas were removed, values written in millions were divided by 1,000 so that every value uses the same unit, and negative values were kept as negatives. After the conversion no missing values were left in any column.

Outliers were checked with the IQR method. They were kept in the data because they are real companies, such as the very large firms at the top of the list, but they are the reason many distributions look so skewed.

Column	Outliers found	Share of 2,001	Comment
Sales	212	10.6%	A few giants above \$400B
Profit	242	12.1%	Includes both big profits and big losses
Assets	241	12.0%	Led by large banks such as ICBC
Market Value	198	9.9%	About seven above \$1T

Seven new features were then created to compare companies fairly regardless of their size:

- Profit Margin (%) = Profit / Sales x 100
- Return on Assets (%) = Profit / Assets x 100
- Asset Turnover = Sales / Assets
- Market Value to Sales and Market Value to Assets
- Profit per Asset
- Companies in Country, the number of listed companies in each company's country

3. Key Performance Indicators

The table below summarises the main indicators of the whole group of companies. Orders, customers and growth rate from the task list were not possible here because the dataset has one row per company for a single year and no customer or date information.

KPI	Value	Unit
Total Sales	51,703.19	\$ billion
Total Profit	4,499.55	\$ billion
Total Assets	238,301.70	\$ billion
Total Market Value	88,453.74	\$ billion
Average Sales per Company	25.84	\$ billion
Average Profit per Company	2.25	\$ billion
Overall Profit Margin	8.70	%
Average Return on Assets	4.65	%
Number of Companies	2,001	companies
Number of Countries	60	countries
Profitable Companies	1,850	companies
Loss-Making Companies	151	companies

4. Exploratory Data Analysis

The notebook contains 15 visualizations. The most important ones are shown here, grouped by what they tell us.

4.1 How big are the companies?

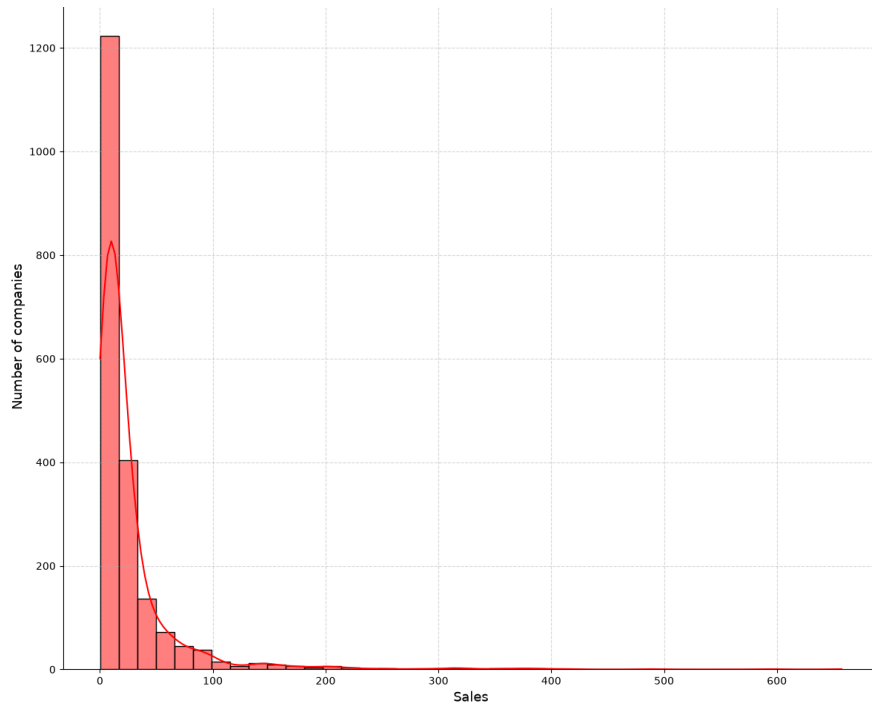


Figure 1: Distribution of company sales (\$ billion)

Most companies are small compared with the giants. Roughly 60% have sales under about \$16 billion, and the distribution has a very long tail towards the right. Profit and market value show the same shape.

4.2 Who leads?

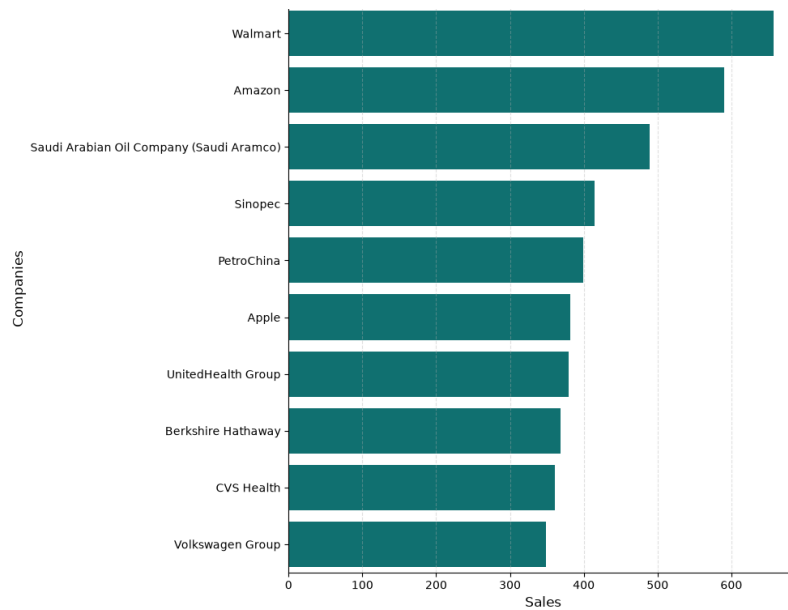


Figure 2: Top 10 companies by sales (\$ billion)

Walmart and Amazon are well ahead of the rest. Three energy companies (Saudi Aramco, Sinopec and PetroChina) and two health companies (UnitedHealth and CVS Health) also make the top 10.

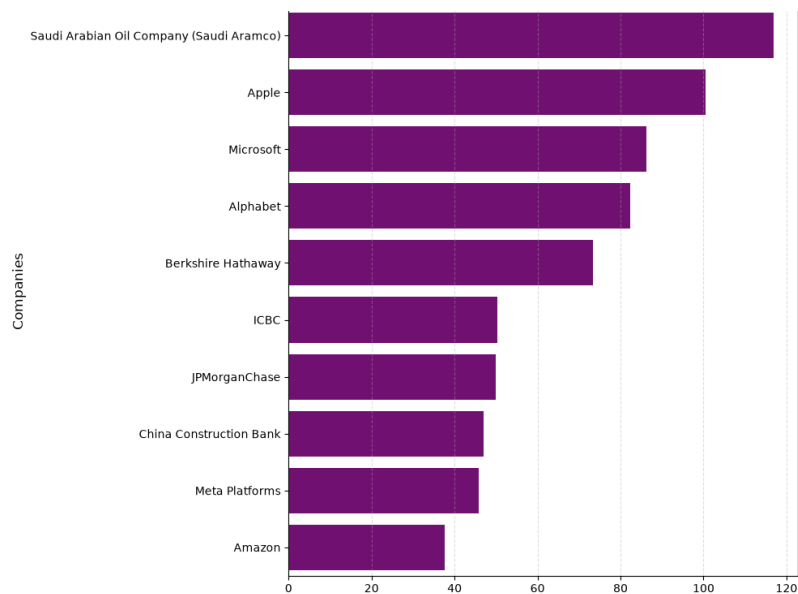


Figure 3: Top 10 companies by profit (\$ billion)

The profit ranking looks very different from the sales ranking. Saudi Aramco, Apple, Microsoft and Alphabet lead, and Walmart is not in the list at all. Three of the ten are banks (ICBC, JPMorgan Chase and China Construction Bank).

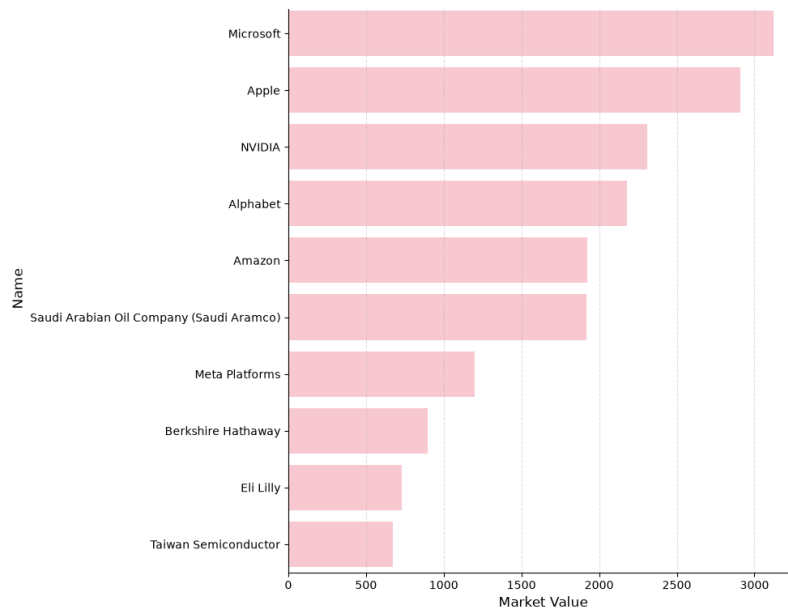


Figure 4: Top 10 companies by market value (\$ billion)

Technology companies take most of the top places. NVIDIA ranks third in market value even though it is not among the ten most profitable companies.

4.3 Where are the companies?

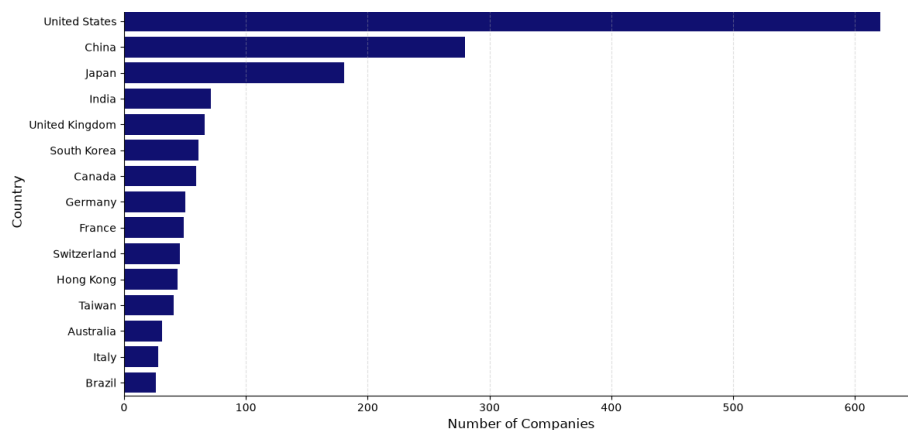


Figure 5: Top 15 countries by number of companies

The United States has 621 companies, China 280 and Japan about 180. Together they account for just over half of the list.

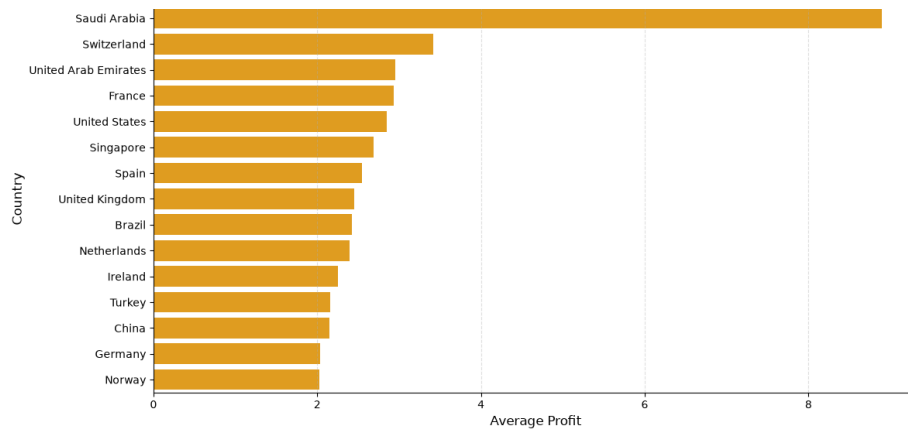


Figure 6: Top 15 countries by average company profit (\$ billion)

Saudi Arabia stands out at about \$8.9 billion, but it has only 16 companies and one of them is Aramco. Switzerland, the United Arab Emirates, France and the United States follow at around \$3 billion.

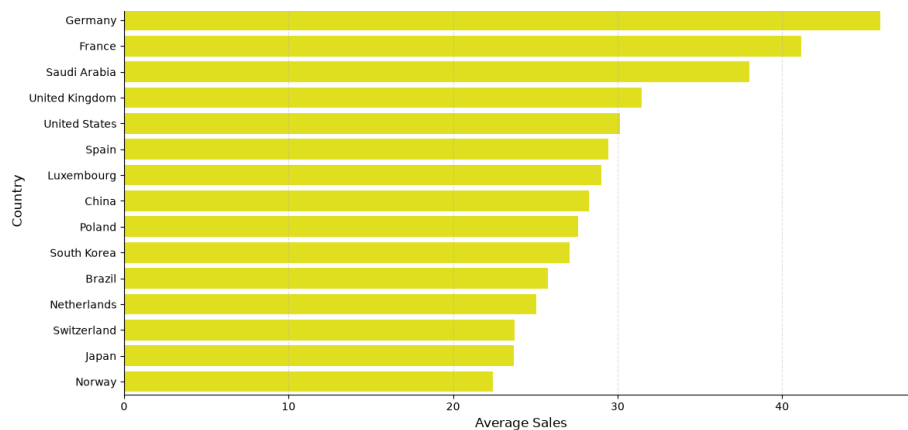


Figure 7: Top 15 countries by average sales (\$ billion)

Germany and France have the highest average sales. Their ranking is different from the profit chart, so high sales and high profit do not always go together.

4.4 How do the numbers relate to each other?

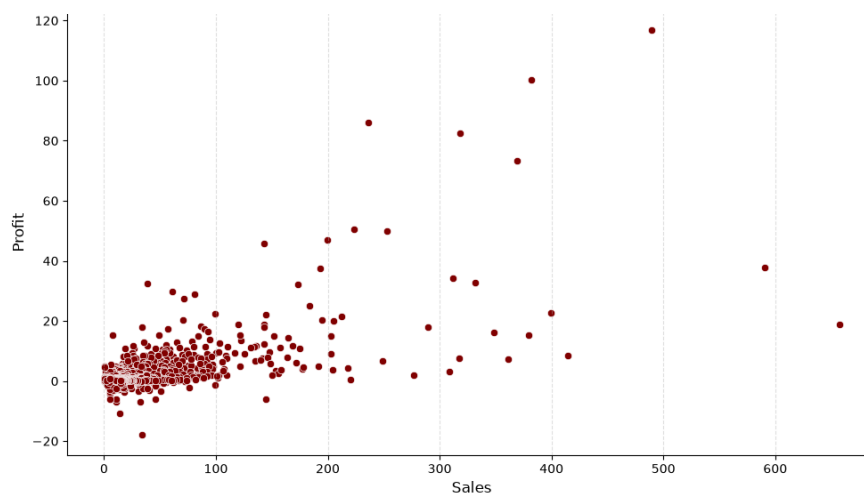


Figure 8: Sales vs profit (\$ billion)

There is a positive relationship, but with a lot of spread. Some companies with sales below \$100 billion earn more than companies with sales above \$300 billion, and Walmart sits far to the right with a low profit.

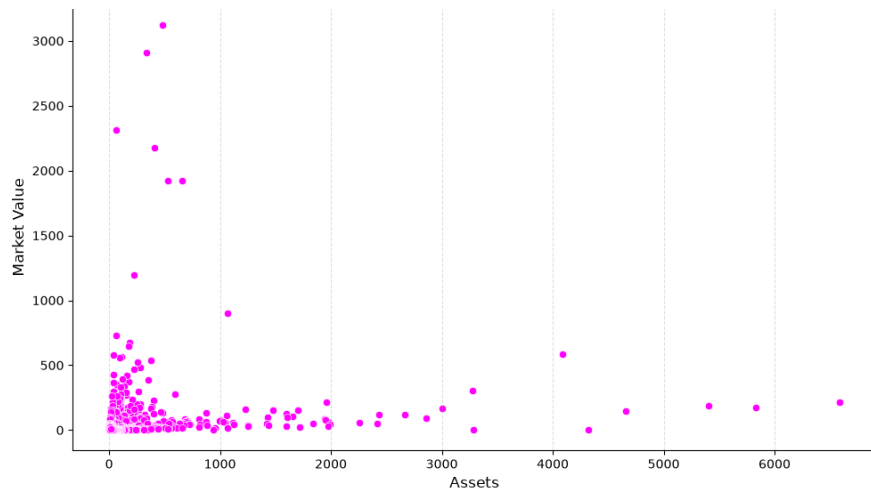


Figure 9: Assets vs market value (\$ billion)

Companies with the highest assets have only modest market values, while the most valuable companies have relatively small assets. Assets alone say little about how the market values a company.

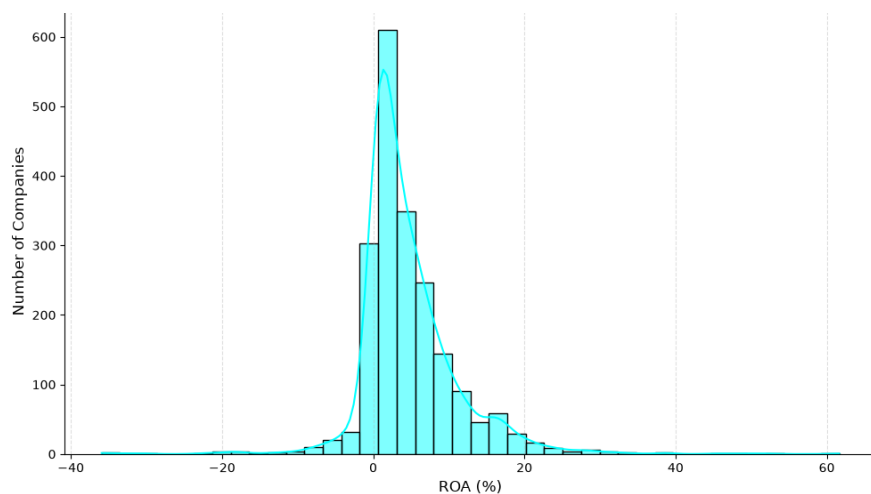


Figure 10: Distribution of return on assets (%)

ROA is centred at a low positive value, with most companies between 0% and 10%. The average is 4.65%.

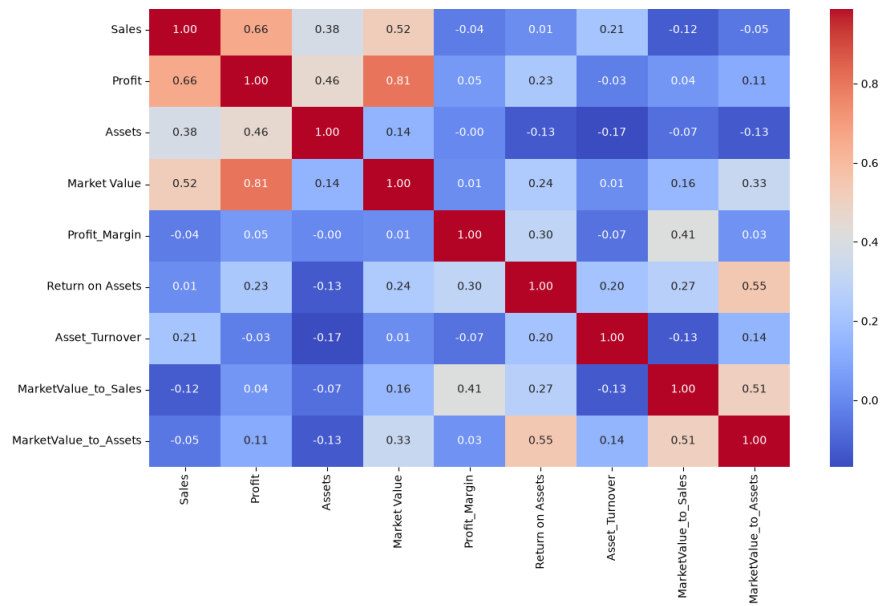


Figure 11: Correlation matrix of the financial features

Market value has the strongest link with profit (0.81), followed by sales (0.66). The ratio columns are mostly weakly related to the raw amounts, which is expected since they remove the effect of size.

5. Insights

The points below come directly from the charts and KPIs above. Values read from charts are approximate.

1. Together the 2,001 companies made about \$4.5 trillion in profit on \$51.7 trillion in sales, an overall profit margin of 8.70%. In simple terms, they kept about \$8.70 of every \$100 they sold.
2. Company size is very uneven. In the sales histogram, roughly 1,200 of the 2,001 companies (about 60%) sit in the lowest bar, under roughly \$16 billion in sales, while a few companies are above \$400 billion. The IQR method flagged 212 outliers in Sales, 242 in Profit, 241 in Assets and 198 in Market Value, which is around 10 to 12% of the data in each column.
3. Walmart is the largest company by sales at roughly \$660 billion, followed by Amazon at roughly \$590 billion and Saudi Aramco at \$489.1 billion.
4. Big sales do not guarantee big profit. Walmart earns only about \$19 billion on its \$660 billion in sales, a margin of around 3%, and it does not appear in the top 10 by profit. Amazon is second by sales but tenth by profit, at about \$38 billion.
5. Saudi Aramco is the most profitable company in the dataset at \$116.9 billion, a margin of 23.9% on its sales. Apple (about \$100 billion profit on about \$382 billion sales) is second, with a margin of roughly 26%.
6. Technology companies dominate market value. Microsoft, Apple, NVIDIA, Alphabet and Amazon take the top five places, and Saudi Aramco is close behind Amazon at \$1,919.3 billion. NVIDIA is third by market value (about \$2.3 trillion) but is not in the top 10 by profit, which suggests its value reflects expected growth more than current earnings.
7. Market value is concentrated in very few firms. The ten largest companies by market value add up to roughly a fifth of the total market value of all 2,001 companies, which is about \$88.5 trillion.
8. The United States has 621 companies (31% of the list), China has 280 and Japan has about 180. These three countries together hold just over half of all companies, and India, the United Kingdom and South Korea come next with 60 to 70 each.
9. Saudi Arabia has the highest average profit per company at about \$8.9 billion, but it has only 16 companies in the list and one of them is Aramco, so the average is driven by a single firm. Switzerland (about \$3.4 billion) and the United States (about \$2.85 billion) are better examples of broad strength.
10. Germany has the highest average sales per company (about \$46 billion), followed by France (about \$41 billion) and Saudi Arabia (about \$38 billion). However, Germany's average profit is only around \$2 billion, near the bottom of the top-15 profit list, so high sales do not always mean high profit at country level either.
11. Profit is strongly linked to market value (correlation 0.81) and to sales (0.66). Its link with assets is weaker (0.46), and the link between market value and assets is only 0.14.

12. Having a lot of assets does not make a company valuable in the market. The largest asset values belong to banks such as ICBC (\$6,586 billion) and JPMorgan Chase (\$4,090.7 billion), yet ICBC's market value is only \$215.2 billion, about 3% of its assets.
13. Return on assets is clustered at low values. Most companies fall between 0% and 10%, the peak is at about 1 to 3%, and the average is 4.65%. There is a long right tail up to about 60% and a small left tail down to about -36%.
14. About 92.5% of the companies (1,850) made a profit and 151 (7.5%) made a loss. The biggest single loss in the data is roughly \$18 billion.
15. Profit margin has extreme values, reaching about 3,600% for at least one company. Values like this usually come from a company with very small sales but a large profit, and they should be treated as outliers or checked before margin is used in any further analysis.
16. A simple linear regression using Sales, Assets and Market Value explains about 67% of the variation in profit (R-squared 0.673). The average prediction error is \$1.13 billion, which is about half of the average profit of \$2.25 billion, so the model is useful but not precise.

6. Recommendations

Each recommendation is based on one or more of the insights above, shown in brackets at the end.

1. Do not rank companies by sales alone. Walmart and Amazon lead in sales but are far behind in profit, so profit margin and return on assets should be shown next to sales whenever companies are compared (Insights 3 and 4).
2. Add an industry column to the dataset and compare each company with similar companies. Margins range from about 3% for Walmart to more than 20% for Apple and Aramco, so one margin target for everyone would be misleading (Insights 4 and 5).
3. Use medians and a minimum company count when comparing countries. Saudi Arabia tops the average profit chart with only 16 companies, so any country with very few companies should be reported with a warning or left out of the ranking (Insight 9).
4. Keep a watchlist of the 151 loss-making companies and review their ROA and profit every year. A company that is both loss-making and low in ROA deserves closer attention than one with a single bad year (Insights 13 and 14).
5. Do not base conclusions only on the United States, China and Japan. They hold just over half of the companies, so country-level benchmarks should also include the next group such as India, the United Kingdom, South Korea and Canada (Insight 8).
6. Use market value as a rough signal of earning power, since it moves closely with profit (0.81), but check the exceptions. Companies like NVIDIA, which rank high in market value but not in profit, need extra analysis before being treated as comparable to the rest (Insights 6 and 11).
7. Judge banks and other asset-heavy companies by ROA and the market-value-to-assets ratio, not by the size of their assets, because large assets did not lead to high market value in the data (Insight 12).

8. Split the companies into size groups (for example large, medium and small by sales) and analyse each group separately. About 60% of companies are small compared with a few giants, so one overall average hides what is typical (Insights 2 and 7).
9. Check and clean the extreme profit margin values before using margin in dashboards or models. Cap the values or analyse those few companies separately so they do not distort averages (Insight 15).
10. Improve the prediction model by applying a log transformation to the skewed columns, adding ratio features, and testing a Random Forest alongside linear regression. The current R-squared of 0.673 gives a clear baseline to beat (Insight 16).

7. Bonus: Predicting Company Profit

As the bonus part, a Linear Regression model was trained to predict a company's profit from its sales, assets and market value. The data was split 80% for training and 20% for testing (random state 42). Profit-based ratios were left out of the inputs on purpose, because they are calculated from profit and would give the model the answer.

Metric	Result	Meaning
MAE	1.13	Average error of about \$1.13 billion
RMSE	2.28	Larger errors count more; about \$2.28 billion
R-squared	0.673	About 67% of the variation in profit is explained

The model captures the main pattern, which fits the correlation results where market value and sales are both strongly related to profit. The errors are still large compared with the average profit of \$2.25 billion, mainly because a few very large companies pull the straight line away from the many small ones. A log transformation or a tree-based model would probably do better.

8. Limitations and Next Steps

- The data covers one year only, so trends and growth rates cannot be calculated.
- There is no industry column, so companies in very different businesses are compared with each other.
- Values taken from charts, such as the top-10 figures, are approximate. Exact values can be read from the notebook.
- The interactive dashboard with filters (Part 4 of the task) is still to be built in Power BI or Tableau using the cleaned CSV saved from the notebook.